# Explicit Hand-Shape Features for ASL Letter Recognition From Categorical Descriptions to Continuous Joint Angles

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## Abstract

Can descriptions that help a person distinguish hand shapes also improve a letter recognizer? Informal webcam confusion among C/D/O and U/V/R motivated this study of explicit hand-shape features added to a Shin-based reference. I compared categorical finger descriptions, workbook-based agreement scores and continuous joint measurements using SVM and LightGBM. All main comparisons used the same 60,251 accepted landmark detections from Kaggle ASL Alphabet and five grouped validation folds. For SVM, adding the full feature package did not improve accuracy (97.628% versus 97.637%). Adding only the joint measurements to the reference reached 97.748%, with macro-F1 increasing from 0.97386 to 0.97524: 67 additional correct predictions. Benefits differed by letter, and some letters regressed. Ablation, grouped permutation and bounded weighting experiments supported a qualified conclusion: the representation matters more than simply supplying more features. The observed angle benefit is small and specific to this evaluation; it does not establish better recognition of unseen people or reliable webcam use.

## 1. Brief Introduction

American Sign Language (ASL) fingerspelling recognition can be approached by locating hand landmarks and classifying their arrangement. This project concerns individual letter images, rather than continuous signing or the movement required for letters such as J and Z. Its practical motivation came from a webcam recognizer: I could intend the same letter but receive a different prediction after changing my hand position. I noticed C/D/O and U/V/R confusions. Those observations were informal and motivated questions; they did not measure error rates.

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Interim course project report. Not a conference publication.

The recognizer followed the geometric approach of Shin et al. (2021). Instead of learning directly from image pixels, it describes distances and directions between hand points. This prompted an intuitive question: could explicit descriptions such as “the middle finger is bent”, “these fingers cross” or “the thumb touches the index finger” make the distinction easier?

I developed a workbook of expected hand features and a tool to inspect the measured hand. I then compared broad bending categories with numerical joint measurements. Figure 1 summarizes that investigation. The contribution is an empirical comparison of these representations, supported by feature-removal tests, model comparisons and letter-level error analysis; it is not a new classification algorithm.

The main research question is: *To what extent do explicit hand-shape features improve ASL letter recognition over a Shin-based reference?* Three questions make it testable:

1. Does the categorical/workbook feature package improve the reference, and does the classifier choice affect the outcome?
2. Are continuous joint measurements more useful additions than the broader feature package?
3. Which letters benefit or regress, and how consistent is the observed improvement?

The following sections connect these questions to prior work, explain the comparisons, report their outcomes and discuss their limits. Later webcam and augmentation experiments are supporting diagnostics, not additional requirements for answering the main question. Technical details and complete results are in the separate appendix.

## 2. Related Works

Shin et al. (2021) represent hand landmarks through pairwise distances and directions and compare SVM with LightGBM. Their approach provides the reference representation here. This study asks whether more explicit hand descriptions help when that geometric information is al-




*Figure 1.* The research sequence. Webcam observations motivated feature design; the primary evidence comes from matched dataset comparisons and analysis of both improvements and regressions.

ready present. Additional features may reorganize existing information rather than supply independent evidence.

MediaPipe Hands estimates a hand skeleton from an RGB image (Zhang et al., 2020). The classifier therefore receives measurements produced by another model. Describing a feature in meaningful terms does not guarantee that the landmark estimate measures it accurately. In contrast, the colour-and-depth approach of Pugeault & Bowden (2011) uses additional sensing information. The present work retains ordinary RGB input and investigates representations of its estimated landmarks.

LightGBM learns boosted decision trees (Ke et al., 2017). Its split-based decisions provide a plausible alternative for mixed hand measurements, while an RBF SVM separates examples through similarity in feature space. Comparing both tests whether a feature package's usefulness depends on the learner. The tested fixed profiles do not establish which algorithm would win after equally extensive optimization.

Interpretability also needs care. Fisher et al. (2019) show that variable importance depends on the fitted model. Accordingly, this study distinguishes shuffling a feature group to test model reliance from removing it and retraining to test added value. Finally, repeatedly selecting models using the same evaluation can produce optimism (Cawley & Talbot, 2010). The experiments below are therefore reported as exploratory comparisons, with their scope separated from final-test claims.

## 3. Methods

### 3.1. From Observations to Representations

The reference contains 820 Shin-style measurements: 190 normalized distances and 630 directions relative to the image axes. Those directions are not finger-joint bending angles. For the additions, I used three representations:

- **Categories:** 70 one-hot inputs describe bending, spread, suspected crossing, thumb shape and contact, including unknown states.
- **Workbook scores:** 26 inputs measure agreement with the expected features of each candidate letter. Slash-separated workbook values are allowed alternatives. Every candidate score is computed without reading the image's true label.
- **Joint measurements:** 14 internal angles and a signed thumb-MCP deviation, giving 15 continuous inputs. MCP measurements are landmark-based proxies rather than clinical joint measurements.

A straight internal angle approaches $180°$; bending reduces it. Categories compress a range of measurements into a label, while continuous values retain finer variation. All additions are derived from landmarks. Their possible advantage is a useful representation for learning, not a new sensor. Main experiments add inputs to the reference; "joint angles" does not mean replacing Shin with an angle-only classifier.

### 3.2. Models and Checks

The main learner is an RBF SVM with $C = 10$, `gamma=scale` and balanced classes. It balances separation against classification errors. LightGBM uses 100 trees, 31 leaves, learning rate 0.1 and minimum child size 20. We retain these project profiles across representation comparisons. Median imputation and standardization of angle inputs are learned only from the fitting partition. Baseline measurements and categorical/workbook inputs pass through unchanged. The appendix gives the objectives, formulas and exact schemas.

To investigate contribution, I removed feature groups and retrained, and separately shuffled groups in held-out inputs of a fitted full-package model. Shuffling used five repeats and up to 100 images per letter per fold. Correlated features can compensate for one another, so these reliance measures are neither causal effects nor additive percentages of importance.

Weighting experiments adjusted the influence of feature blocks. Bounded searches used five outer folds and three inner group folds; inner searches were capped at 150 im-

ages per letter, with final fits using all outer-training images. This limits tuning cost. It does not remove selection effects from the broader, exploratory research sequence.

## 4. Experimental Results

### 4.1. Dataset and Evaluation

Kaggle ASL Alphabet supplies 78,000 training images, 3,000 for each A–Z label. Only 60,251 images with exactly one accepted hand detection enter the comparisons: 77.25% coverage. Consequently, classification scores are conditional on successful landmark extraction. The small supplied test set duplicates training images and was excluded. Historical landmark records do not retain the confidence thresholds; the intended default extraction settings cannot be independently confirmed.

Every main comparison uses the same accepted images and frozen five group folds. Adjacent image numbers are grouped into six filename blocks, distributed across the folds. Each image receives one prediction from a model that was not fitted on it. These groups are not verified people; possible similarity across blocks limits claims about new signers. Accuracy is pooled over all held-out predictions. Macro-F1 averages the 26 letter F1-scores, giving equal weight to each letter and accounting for false positives as well as missed examples. No untouched final classification test is available.

### 4.2. RQ1: Does the Broader Feature Package Help?

Table 1 gives the matched results. For SVM, all 111 additions together reach 97.628% accuracy, slightly below the 97.637% reference. LightGBM improves from 97.097% to 97.164% with that package, but remains below the tested SVM configurations. Thus the package's effect depends on the learner, and it does not consistently improve the reference.

Feature removal clarifies the SVM result. Removing the 70 categories from the full package improves accuracy to 97.744%, the angles-plus-workbook row. Removing angles instead leaves categories and workbook scores and reaches 97.593%. These comparisons support the continuous representation in this setting. They do not isolate categories alone without workbook scores, or prove that categorical descriptions are inherently unhelpful.

### 4.3. RQ2: Are Continuous Measurements More Useful?

Adding the 15 joint measurements alone to the reference produces the strongest simple SVM representation in Table 1: 97.748% accuracy and 0.97524 macro-F1, versus 97.637% and 0.97386. "Alone" here concerns the additions, not the full model input. The workbook scores add no further benefit to that configuration. For LightGBM, the angle addition instead reduces accuracy to 97.017%. The result is therefore specific to the tested learner and preprocessing.

Grouped permutation gives complementary evidence. In the full-package SVM, shuffling joint measurements reduces macro-F1 by an average 0.570 percentage points, versus 0.049 for categories and 0.008 for workbook scores. This indicates greater reliance on the angles in that fitted model; it is not the amount by which adding them improves the reference. The removal and matched-addition results are needed for that claim. Broad feature weighting did not yield a convincing additional overall benefit under the bounded search.

### 4.4. RQ3: Who Benefits, and How Stable Is the Gain?

The angle addition fixes 143 formerly incorrect images and loses 76 formerly correct ones: a net gain of 67 out of 60,251, or 0.1112 percentage points. The improvement occurs in all five outer folds and all six filename blocks, but is small. A block-resampling sensitivity analysis gives a central range of 0.0374–0.1947 percentage points. With only six non-person blocks, overlapping fits and exploratory model selection, this is not a valid independent-test confidence interval or a confirmatory significance result (Appendix D).

Figure 2 and Table 2 show why the pooled gain needs qualification. M gains 31 correct images, while O loses seven. N loses two true-letter recoveries yet improves in F1, because false-positive behaviour also changes. Different metrics therefore answer different questions. The angle model still has 142 R→U, 141 N→M, 102 M→N and 41 D→O errors. These residual patterns matter more than selecting only favourable letters.

A targeted index-joint weighting study explored one suspected D/O mechanism. Nested weighting reduced O→D from 19 to 15 errors, but increased N→M from 141 to 146 and lowered overall accuracy to 97.739%. Nested PIP-only weighting reached 97.753%, effectively the same overall level as 97.748%. The selected weights differed between folds. These checks do not support a universal instruction to give the index finger greater influence.

### 4.5. Supporting Webcam Diagnostics

A later guided development round retained all 27 planned C/D/O attempts with predictions hidden. The reference and angle models each recognized 17/27. Small tilt augmentation recognized 18/27; C-only tilt did not improve the predictions. Later 2D-angle and G/H-specific investigations likewise did not establish a general solution. Those results

*Table 1.* Pooled five-fold results on the same 60,251 accepted images. Every row retains the 820 reference inputs. "All additions" adds 70 category indicators, 26 workbook scores and 15 joint measurements. These are matched exploratory results, not independent final-test scores.

| | | SVM | | LightGBM | |
|---|---|---|---|---|---|
| Representation | Inputs | Accuracy (%) | Macro-F1 | Accuracy (%) | Macro-F1 |
| Reference | 820 | 97.637 | 0.97386 | 97.097 | 0.96784 |
| All additions | 931 | 97.628 | 0.97372 | 97.164 | 0.96860 |
| Joint angles | 835 | 97.748 | 0.97524 | 97.017 | 0.96692 |
| Workbook scores | 846 | 97.648 | 0.97400 | 97.119 | 0.96815 |
| Angles + workbook scores | 861 | 97.744 | 0.97515 | 97.129 | 0.96824 |

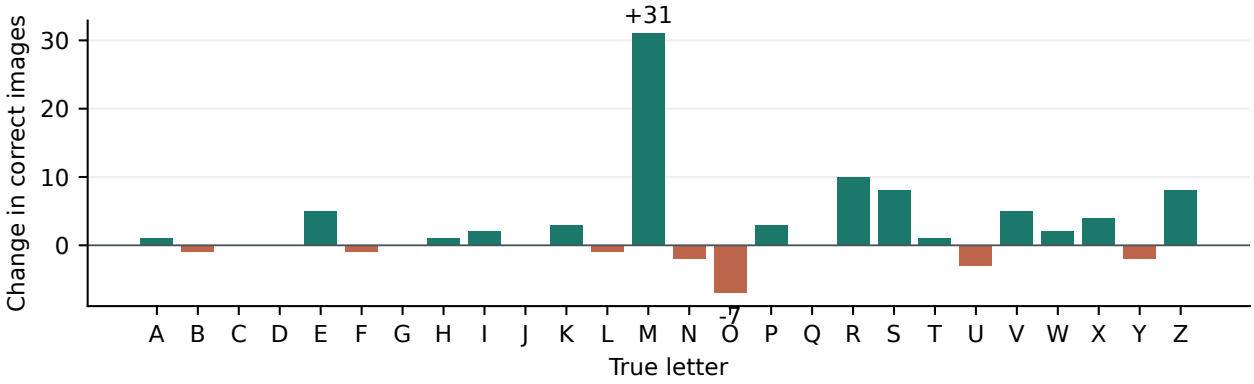



*Figure 2.* Change in correct true-letter images after adding continuous measurements to the SVM reference, using the same images and folds. Green bars show gains and orange bars losses. Their sum is +67; the gain is not uniform across letters. Full precision, recall, F1 and confusion matrices are in Appendices A–B.

*Table 2.* Selected letter effects; F1 is a percentage and $\Delta n$ is the change in correctly recovered true-letter images. The appendix includes every letter.

| Letter | Images | Base F1 | Angles F1 | $\Delta n$ |
|---|---|---|---|---|
| M | 1602 | 89.37 | 90.51 | +31 |
| N | 1280 | 88.51 | 89.61 | -2 |
| O | 2264 | 97.52 | 97.30 | -7 |
| R | 2543 | 94.81 | 95.33 | +10 |
| S | 2559 | 97.48 | 97.77 | +8 |
| Z | 2343 | 98.26 | 98.37 | +8 |

are retained in Appendix I–J, outside the primary feature comparison.

These observations illustrate a limit, not a new webcam accuracy claim. For example, one selected G had middle-PIP flexion consistent with training G but was still H; another H with visible palm was G. Palm/back viewpoint, other measurements and learned decision boundaries may all contribute. Selected errors cannot establish the frequency or main cause of failure, and projected image angles do not prove that an estimated 3D angle is wrong.

## 5. Discussion

The answer to the main question is qualified: explicit handshape features do not automatically improve the recognizer. The full package gives no SVM advantage; continuous measurements give a small positive dataset result when added to the reference. This is a useful negative-and-positive comparison, rather than evidence that more features are always better.

A plausible interpretation is that joint measurements reorganize geometry into a useful form for the SVM, whereas broad categories discard detail and some additions are redundant. The experiments support the observed performance differences, not that mechanism as a proven cause. The LightGBM results also show that an informative representation for one learner need not help another. These findings concern feature design, not a new algorithm or validated anatomy.

Several limitations bound the answer. The workbook and its relaxations were developed exploratorily, not nested inside all later evaluations. Multiple representations were inspected on the same folds, despite nested tuning within particular searches. The strongest observed result should therefore not be treated as an unbiased estimate of a selected final system. Fixed classifier profiles also limit conclusions about algorithm superiority. Filename-block grouping cannot establish signer-independent performance, and detection filtering excludes difficult images before classification.

The source article reports 87.60% ASL Alphabet accuracy, not a matching F1 score. Our filtering, split and training settings differ, so the local 97% scores are neither an exact reproduction nor evidence of outperforming that article. Live examples further show that a strong conditional dataset score does not guarantee reliable use under different capture conditions. No evidence here establishes that back-of-hand views are systematically worse, or that the early U/V/R observations have been solved.

The report and much of the software were AI-generated. AI-assisted code inspection, unit tests and reconciliation with stored predictions were used in an attempt to verify the implementation. These checks do not constitute independent human code review, linguistic review of the workbook or validation of estimated anatomy. The author should verify the interpretation and references before submission. Participants should consent to webcam collection; saved images remain local by default. Single-frame letter recognition should not be presented as full sign-language understanding or a replacement for qualified interpretation.

## 6. Conclusions

Within the shared five-fold accepted-image evaluation, adding continuous joint measurements to the Shin-based SVM raises accuracy from 97.637% to 97.748% and macro-F1 from 0.97386 to 0.97524. The gain is small, depends on representation and learner, and includes letter-level regressions. Broad categorical/workbook additions and increased weighting offer no consistent alternative advantage under the tested settings.

The main contribution is showing which explicit representations help in this evaluation, using matched comparisons, ablations and error analysis, while separating dataset gains from webcam reliability. Future work should predefine one comparison and evaluate it on a new, balanced signer/session-separated collection. Additional viewpoint data would test generalization; it is not required to reinterpret the current modest feature-comparison result as a solved webcam problem.

## References


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